

MANDAL II, Norway

WMO: 014300

Lat: 58.0284N

Lon: 7.4527E

Elev: 138

StdP: 99.68

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.1	(c) -8.1	(d) -14.4	(e) 1.1	(f) -8.4	(g) -12.1	(h) 1.3	(i) -6.0	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.7	(c) 23.6	(d) 16.5	(e) 21.9	(f) 15.8	(g) 20.5	(h) 15.3	(i) 18.0	(j) 21.2	(k) 17.3	(l) 20.1	(m) 16.5	(n) 19.2	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.2	19.1	16.1	11.7	18.6	15.3	11.1	17.8	51.4	21.3	49.1	20.2	46.8	19.2	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -11.3	(f) 25.8	(g) 3.0	(h) 1.3	(i) -13.4	(j) 26.7	(k) -15.1	(l) 27.4	(m) -16.8	(n) 28.2	(o) -18.9	(p) 29.1		
(4) N/A	(5) N/A	(6) N/A	(e) -11.9	(f) 19.1	(g) 2.8	(h) 1.1	(i) -13.9	(j) 19.9	(k) -15.5	(l) 20.5	(m) -17.1	(n) 21.1	(o) -19.1	(p) 21.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	0.8	0.9	3.4	6.6	10.7	14.1	16.5	15.6	13.1	8.9	5.1	2.3
	DBStd	6.37	3.84	3.86	3.25	2.88	3.60	2.13	2.32	1.66	2.13	2.63	3.48	4.64
	HDD10.0	1328	286	255	205	109	33	0	0	0	3	52	147	238
	HDD18.3	3712	544	488	463	353	237	129	67	87	157	294	396	497
	CDD10.0	676	0	0	1	6	56	123	202	173	97	17	1	0
	CDD18.3	16	0	0	0	0	1	2	11	2	0	0	0	0
	CDH23.3	57	0	0	0	0	3	7	46	1	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	1627	190	133	105	85	78	95	100	133	153	189	181	175
	PrecMax	2316	424	340	239	142	161	211	244	242	302	339	506	340
	PrecMin	905	17	15	14	24	7	8	10	13	81	50	68	31
	PrecStd	304	88	87	49	34	42	53	55	53	51	81	96	83
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	9.7	14.5	17.0	23.7	24.3	26.7	22.7	20.2	15.8	12.4	10.4
		MCWB	7.0	6.5	8.8	9.5	15.4	16.2	17.7	17.5	15.6	13.0	11.0	9.0
	2%	DB	7.8	7.7	10.9	15.1	21.7	21.9	24.7	20.9	18.5	14.5	11.3	9.6
		MCWB	6.4	5.8	6.8	8.8	14.2	14.7	17.5	16.2	14.7	12.5	9.9	8.4
	5%	DB	7.0	6.6	9.1	13.2	19.3	20.2	22.7	20.0	17.5	13.6	10.6	9.0
		MCWB	5.8	5.0	5.7	8.2	13.5	14.0	16.7	15.8	14.4	11.7	9.2	7.8
	10%	DB	6.3	5.9	7.7	11.3	17.0	18.7	21.0	19.1	16.7	12.9	9.9	8.1
		MCWB	5.0	4.6	5.2	7.1	12.2	13.3	16.1	15.1	14.1	10.8	8.6	6.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	7.9	9.2	11.2	16.3	17.5	19.5	18.7	17.3	14.1	11.4	9.5
		MCDB	8.4	8.9	12.8	14.1	21.9	22.6	23.5	21.0	18.4	15.1	12.1	10.0
	2%	WB	6.7	6.4	7.7	9.8	15.1	16.0	18.4	17.7	16.3	13.2	10.3	8.7
		MCDB	7.5	7.0	10.0	13.3	20.3	19.9	22.3	19.7	17.5	14.3	11.1	9.4
	5%	WB	6.2	5.6	6.6	8.7	14.0	15.1	17.6	16.9	15.4	12.3	9.5	8.1
		MCDB	7.0	6.3	8.3	12.4	18.3	18.6	21.1	18.8	16.9	13.3	10.4	8.9
	10%	WB	5.2	4.7	5.7	7.6	12.7	14.2	16.9	16.2	14.4	11.5	8.8	7.0
		MCDB	6.2	5.8	7.2	10.6	16.2	17.4	20.3	18.0	16.0	12.5	9.9	8.0
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.9	5.9	6.5	7.2	7.1	6.7	6.4	5.9	5.1	4.3	4.5
		MCDBR	5.2	5.6	8.5	9.1	10.1	10.1	9.3	7.3	7.1	5.5	4.7	4.2
		MCWBR	5.0	4.4	5.4	5.0	5.0	5.2	4.6	4.0	4.4	4.1	4.2	4.3
	5% WB	MCDBR	5.2	5.2	7.1	7.9	8.8	8.3	7.6	5.5	5.4	4.5	4.4	4.5
		MCWBR	5.1	4.6	4.8	4.7	4.8	4.9	4.1	3.9	4.3	4.1	4.3	4.7
Clear-Sky Solar Irradiance	taub	0.300	0.326	0.342	0.379	0.368	0.356	0.372	0.375	0.355	0.335	0.307	0.279	
	taud	2.230	2.320	2.369	2.317	2.422	2.477	2.449	2.437	2.484	2.441	2.293	2.184	
	Ebn at Noon	536	707	806	829	867	883	858	827	789	694	527	438	
	Edh at Noon	49	71	88	108	104	100	101	96	79	62	46	37	
All-Sky Solar Radiation	RadAvg	0.34	0.94	2.30	3.86	5.22	5.85	5.28	4.04	2.62	1.25	0.47	0.23	
	RadStd	0.07	0.19	0.24	0.43	0.61	0.53	0.56	0.33	0.27	0.19	0.10	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-122	-162	N/A	N/A

Nomenclature: See separate page